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RESEARCH ARTICLE

10.1029/2026SW005072

Key Points:

- Operators implemented many of the identified mitigation options but favored low-cost strategies over expensive hardware upgrades
- Mission continuity and revenue concerns often lead satellite and power operators to bypass conservative risk reduction protocols
- Resilience in critical infrastructure is driven by timely forecast integration and information sharing over physical system modifications

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

LaNeve, C. M., Oughton, E. J., Rivera, N., Weigel, R. S., Thomas, D., & Gaunt, C. T. (2026). Space weather mitigation decisions by critical infrastructure operators during solar cycle 25. *Space Weather*, 24, e2026SW005072. <https://doi.org/10.1029/2026SW005072>

Received 18 MAR 2026

Accepted 15 SEP 2026

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Space Weather Mitigation Decisions by Critical Infrastructure Operators During Solar Cycle 25

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Abstract The “Gannon Storm” in May 2024 was the largest space weather event experienced in 20 years, generating auroras at latitudes as low as 35°, visible in Florida and Mexico. Such events pose significant operational challenges for critical infrastructure operators, particularly those managing electricity transmission networks, satellite constellations, and aviation systems. Substantial progress has been made in understanding space weather and the potential exposure of infrastructure assets to severe events. Yet, we have seen very little evaluation of the types of current mitigation strategies used to reduce our shared vulnerability to this activity, motivating this study. First, we filled an important literature gap by undertaking a review of the range of space weather mitigation strategies available. Then, we contacted 313 critical infrastructure operators (50 power, 227 satellite, 36 aviation) for participation in an anonymous online survey and/or interview receiving 55 unique responses (18% response rate). We developed a time-sensitive data set, through conducting qualitative interviews with 33 operators, capturing operational decisions made during peak solar activity. The results identified 91 potential mitigation decisions across the three sectors and found that 74% of these were undertaken. This is one of the first substantial studies of space weather mitigation activities and moves beyond the traditional focus on physical and economic impacts.

Plain Language Summary This study focuses on how power, satellite, and aviation operators handled space weather events over Solar Cycle 25, comparing academic advice with industry actions. While research often suggests large purchases of expensive hardware, or shutting down systems to avoid damage, most companies opted to continue operations to avoid losing money, relying on modeling and alert systems to manage risk. We identify a preparedness gap where larger companies have the ability to harden networks before operations, but smaller ones may choose to accept the risk of the storm due to high costs. Ultimately, this study highlights that instead of only purchasing new equipment, operators identified the need for more accurate, localized, space weather warnings from the government to protect critical infrastructure.

1. Introduction

Space weather arises from many different types of solar phenomena, including solar flares, Coronal Mass Ejections (CMEs), and solar proton events, presenting hazards to assets in orbit, as well as technology and infrastructure on Earth's surface (Xue et al., 2024). The Sun has a cycle of increased and decreased solar activity over approximately 11 years, covering the solar maximum and minimum periods (Cao et al., 2022). The reversal of the Sun's magnetic poles usually occurs during solar maximum, a period of increased activity associated with frequent solar flares and sunspots (Interrante, 2024; Jones et al., 2020). This period of increased activity causes higher chances of disruption associated with space weather, including but not limited to impacts to power grids, satellite operations, and global positioning and navigation (Ciliverd et al., 2025; Kashcheyev et al., 2025; Mal'etckii & Astafyeva, 2024). Unfortunately, we still lack information on how critical infrastructure operators respond to heightened space weather conditions.

The geomagnetic storm of 10 May 2024, named the “Gannon Storm,” was the first time since the 2003 Halloween storms that the US National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC) issued a G5 geomagnetic event warning, the highest rating on the scale (Evans et al., 2024; Horne et al., 2026). Extreme space weather events provide crucial data for modeling and prediction purposes, with many using data from the Carrington superstorm of 1859 and the 2003 Halloween storm for validation (Blake et al., 2021; Ngwira et al., 2014). Therefore, the importance of this event provides strong motivation to collect

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perishable operational decision data from critical infrastructure operators, especially in critical infrastructure sectors essential for society and the economy.

To increase societal resilience to space weather threats, we need to be fully aware of the range of potential mitigation decisions available to, and in current use by, critical infrastructure operators. To do so, we need to collect, visualize, analyze, and document this information, for example, over the solar maximum of the most recent solar cycle. Subsequently, this study considers the following three research questions:

1. What is the current range of mitigation decisions available to critical infrastructure operators to protect against space weather?
2. Which of these decisions were carried out over the current solar maximum (Solar Cycle 25), including the May 2024 “Gannon Storm?”
3. How do the mitigation decisions identified from the literature compare to those made by critical infrastructure operators during recent solar events?

A literature review in the following section leads to the methodology presented in Section 3. The results are then presented and discussed in Section 4, before returning to the research questions in Section 5 to consider wider ramifications. Section 6 provides conclusions.

2. Literature Review

A systematic review of the literature is carried out in this section focusing on three major critical infrastructure sectors: power, satellite, and aviation.

2.1. Electricity Transmission (Power) Infrastructure Impact Mechanisms

Power systems are a “foundational” infrastructure, as their failure cascades into all other sectors (Hall et al., 2016, 2017). Space weather in the form of Geomagnetic Disturbances (GMDs) poses a significant risk to power grids and Extra High Voltage (EHV) transformers (>200 kV) (Albertson & Van Baelen, 1970; Jonas & McCarron, 2015; Oughton et al., 2025; Pirjola & Boteler, 2002). Near-Earth space weather is driven by plasma with an embedded magnetic field continuously emitted from the Sun, which interacts with Earth's internally generated magnetic field. One of the primary drivers of near-Earth space weather is a CME (Chapman, 1918; Liu et al., 2024; E. N. Parker, 1958; Pulkkinen et al., 2025), a high-speed, high-density burst of plasma with a large magnetic field. The interaction between solar plasma and Earth's magnetic field drives energization processes in the magnetosphere, leading to geomagnetic storms and substorms, which are associated with enhanced and variable currents flowing into and out of the ionosphere (Moldwin, 2008; Thomas et al., 2024; Woodroffe, 2019). Time-varying currents in the magnetosphere and ionosphere lead to electromagnetic waves that penetrate into and are reflected by Earth's conducting surface. This results in a time-varying electric field on Earth's surface that drives slowly varying geomagnetically induced currents (GICs) with the significant power in low frequency components, less than 1 Hz, into conductive networks such as power grids and pipelines (Birkeland, 1908; Clilverd et al., 2025; Espinosa et al., 2023; Marshalko et al., 2023; Nakamura et al., 2018). This result is a time-varying electric field on Earth's surface that drives GICs into conductive networks such as power grids and pipelines (Ingham et al., 2023; Molinski, 2002; Ngwira et al., 2023; Oyedokun et al., 2020; Pratscher et al., 2024; Thomas et al., 2024).

Generally, the GICs in a power grid depend on the voltage, configuration, length, local ground resistivity, and spatial direction of the network (Andreyev et al., 2023; Mayer & Stork, 2024). Substations located in areas of high ground resistivity or connected to long transmission lines are exposed to stronger induced geoelectric fields (Barnes et al., 2024; Kazerooni et al., 2017; Oughton et al., 2025). In terms of network configuration, longer and higher voltage transmission lines tend to have higher GICs, with the potential to flow through EHV transformers (>200 kV) (Caraballo et al., 2023; Kazerooni et al., 2017). The low frequency GICs, sometimes referred to as quasi-DC, flowing through EHV power transformers cause saturation of the transformers' magnetic cores, leading to inefficient operation (Ahmadzadeh-Shooshtari & Rezaei-Zare, 2022a; Subritzky et al., 2024). Three main mechanisms of failure from GICs have been identified. First, prolonged saturation can overheat transformer windings, leading to permanent failure (Akbari & Rezaei-Zare, 2023; Akbari et al., 2023; Albert et al., 2022). Second, inefficient power delivery, often referred to as excessive flow of reactive power, can lead to voltage instability and, potentially voltage collapse of the grid (Saleh et al., 2024). Third, harmonics generated by the core

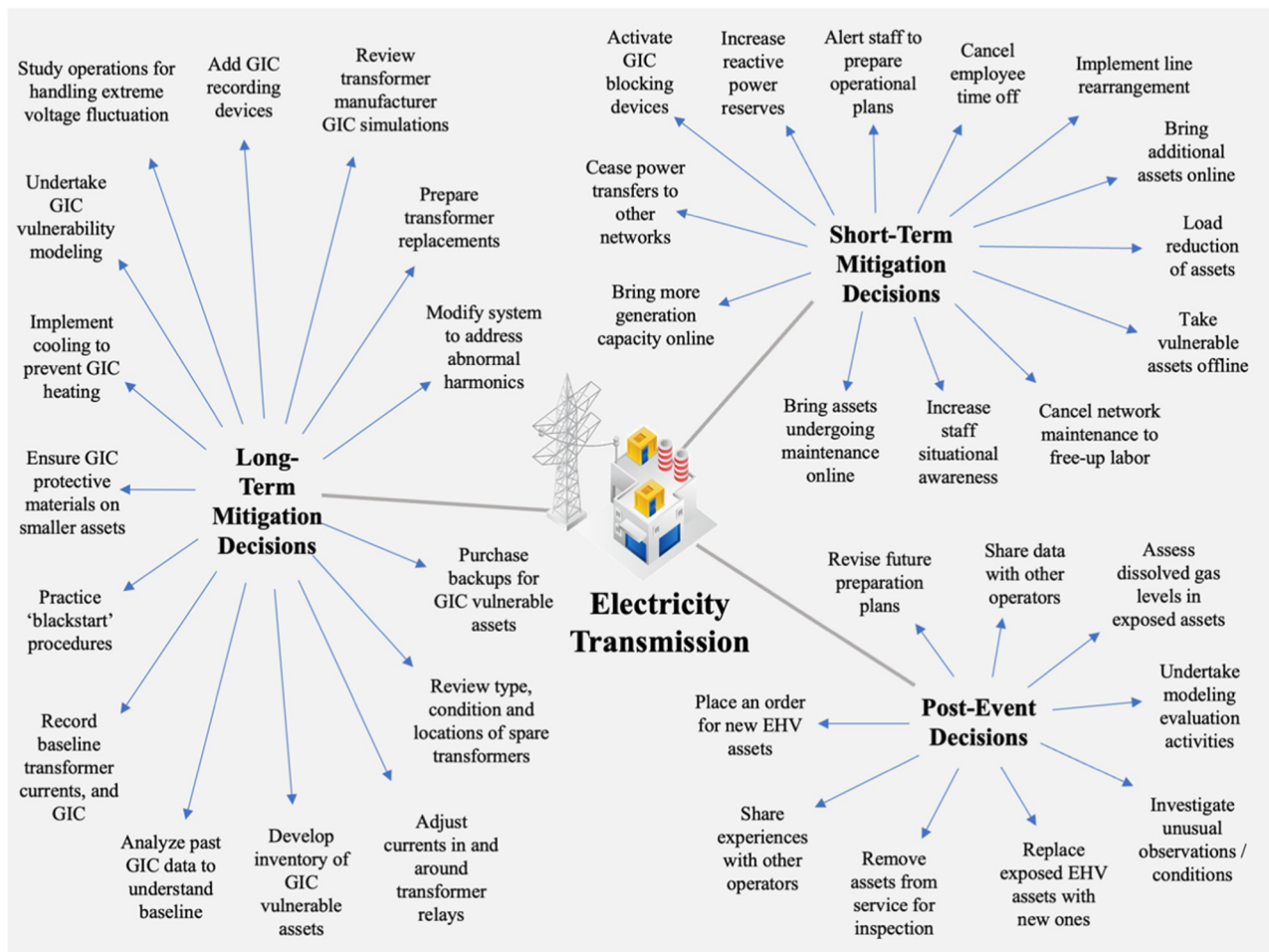


Figure 1. Strategic framework for mitigating GMD-induced risks to Extra High Voltage assets, from infrastructure hardening (long-term) to grid stabilization (short-term) and forensic analysis (post-event).

saturation can affect grid protection systems and operations (Ahmadzadeh-Shoostari & Rezaei-Zare, 2022b; Crack et al., 2024). Isolation of important components, including transformers, lines, and shunt capacitors, from the grid can lead to cascading failure with widespread power outages, as seen during the 1989 Quebec blackout (Boteler, 2019; Oughton et al., 2024).

2.2. Electricity Transmission (Power) Infrastructure Mitigation Options

To address these vulnerabilities, the literature identifies a range of mitigation strategies, categorized by their implementation timeline relative to a GMD event (summarized in Figure 1). Long-term strategies focus on structural resilience and predictive modeling, including hardening hardware through cooling systems and GIC-blocking devices as well as establishing baseline transformer metrics to refine vulnerability assessments (NERC, 2019; U.S. Department of Energy (DOE), 2019).

Short-term operational decisions are implemented when an extreme space weather event is imminent, and a space weather forecast has been issued for an incoming event (Smith et al., 2022). These actions prioritize grid stability through load reduction, staff mobilization, and strategic reconfiguration of transmission lines to manage reactive power reserves (Etchells et al., 2024; Jasiūnas et al., 2021; Mac Manus et al., 2023; NERC, 2012; Patel, 2024). Following an event, post-event recovery emphasizes forensic analysis, such as dissolved gas analysis in EHV assets, and the sharing of observational data to improve future modeling and response accuracy (Caraballo et al., 2025; Gonzalez-Esparza et al., 2024; Song et al., 2022). Following the SWPC warning of the Gannon storm issued in the USA and by the relevant authorities in New Zealand and the UK, some electricity utilities

implemented short term mitigation actions and post-event reviews (Horne et al., 2026; Transpower, 2024). The New Zealand national grid operator Transpower deployed a switching strategy within the network ahead of the Gannon storm, an action that reduced the peak GIC from a projected 200 A to an observed 113 A (Oughton et al., 2026).

2.3. Satellite Impact Mechanisms

Solar activity exerts significant impact on satellites in orbit, including disruption of communications, surface charging of electronics, and changes to orbital drag (Aleshko et al., 2023; Allen et al., 1989; Gabrielse et al., 2022; O'Brien et al., 2022; Yue et al., 2023). Increased solar radiation can also cause Single Event Upsets across electronic components commonly used in constellations (Zhang et al., 2023).

Satellite operators face a unique decision-making environment characterized by high-cadence telemetry data contrasted with sparse environmental ground truth. Unlike terrestrial operators who can physically inspect assets, satellite operators rely on remote diagnostics and partial historical space weather records; this information gap mandates highly confident, actionable forecasts (Boyd et al., 2023; Feng et al., 2023). An operator will prioritize the safety of the satellite equipment, resulting in transferring the asset into safe mode, a low power operating state where all non-essential systems are shut down (Arif, 2025; Qiao et al., 2024; Trimoreau, 2022). Space weather events, such as geomagnetic storms and substorms, can alter the structure of Earth's upper atmosphere, resulting in low Earth orbiting satellites experiencing increased atmospheric drag due to the increased local density of the gas and plasma in their orbital path (Burke, 2018; Fang et al., 2022; Sanchez-Hurtado et al., 2025; Yin et al., 2024). These dynamics are exacerbated by the rapid proliferation of satellite “mega-constellations” (e.g., Starlink) and growing debris fields, which demand rigorous collision avoidance protocols during storm-induced drag events (Kukreja et al., 2025; Osoro et al., 2025; W. E. Parker et al., 2024).

The second primary hazard is that satellites are susceptible to ionizing and non-ionizing radiation, which plays a significant role in satellite reliability. This subject is complex. The dominant degradation source in some NASA missions is proton-induced, which is a mix of Total Ionizing Dose (TID) and non-ionizing Displacement Damage Dose (DDD) (Topper et al., 2017). For surface and internal charging, the focus is on the electron flux. The electron flux is larger than the proton flux, and electrons of the same energy as protons will penetrate further into the spacecraft. The resulting charging is energy dependent, as lower energy electrons charge external surfaces to varying potentials, while higher energy electrons charge internal dielectrics and ungrounded conductors (Horne et al., 2025). Both can result in damage from electrostatic discharge (Garrett & Whittlesey, 2011).

Single Event Effects constitute a third major hazard, manifesting as single instance failures triggered by high-energy particles or other environmental stimuli (Etchells et al., 2024). Elevated fluxes of trapped electrons or other high-energy particles during space weather events can induce transient or permanent damage to satellite systems (Hands et al., 2018; Xue et al., 2024). Given that space weather events can induce all three effects on a satellite, similar mitigation procedures often apply.

2.4. Satellite Mitigation Options

To manage the complex interplay of radiation and atmospheric drag, operators employ a tiered mitigation strategy ranging from hardware hardening to orbital maneuvers (summarized in Figure 2). Long-term strategies focus on design-phase resilience, primarily through radiation shielding evaluation and the implementation of redundancy in onboard electronics and ground-based power supplies (Leahy, 2021; Praks et al., 2023; Shah et al., 2024). Furthermore, longitudinal monitoring of orbital radiation environments allows for the refinement of predictive exposure models and future asset replacement schedules (Camporeale et al., 2022; Kieokaew et al., 2026; Qiao et al., 2024). Monitoring future orbits enables operators to predict which assets will be most exposed and make necessary adjustments (Miteva et al., 2023; Spence et al., 2022).

Short-term mitigation decisions focus on minimizing potential damage and maintaining service continuity. Operators alert staff to prepare operational plans and notify customers of possible service disruptions (Blumenfeld, 2024). At-risk satellites may be maneuvered to safer orbits, while placing satellites into safe mode protects sensitive electronics from excess radiation (Mahmood et al., 2025; Shambaugh, 2025). Increasing situational awareness among staff, canceling employee time off, and ensuring adequate personnel availability by

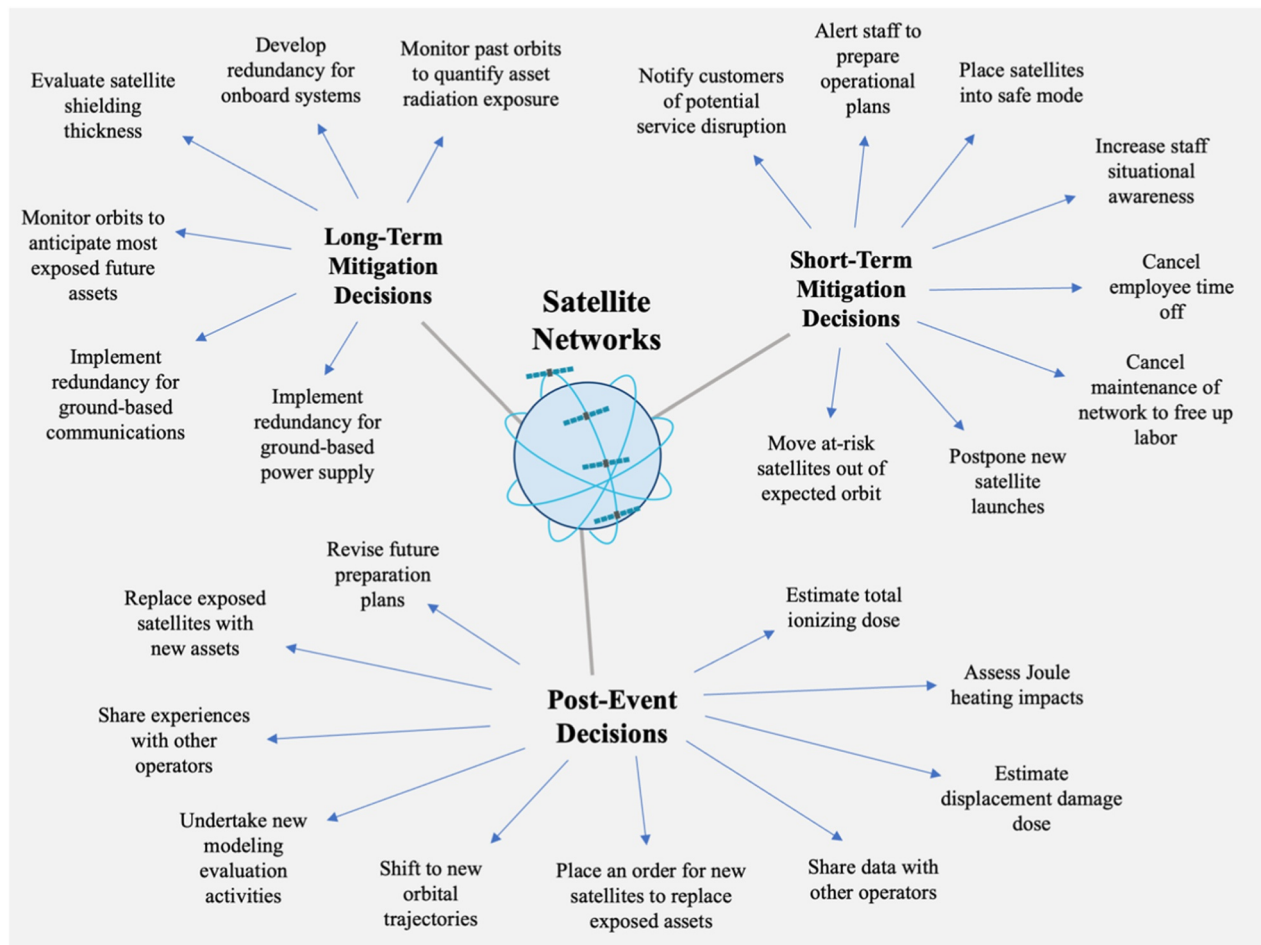


Figure 2. Strategic framework for satellite network resilience, from design-phase hardening (long-term) to orbital management (short-term) and post-event degradation assessment.

postponing maintenance and new satellite launches ensure availability to support event response (Fang et al., 2022). These actions reduce the likelihood of irreversible damage while preserving essential functions.

Post-event mitigation centers on evaluating impacts and restoring operational integrity. This includes estimating absorbed TID to determine component lifetime and performance (Leahy, 2021; Serief & Meguenni, 2025). Damaged satellites may be replaced, and orbital adjustments may be implemented to optimize future performance (Praks et al., 2023). Constellations such as Starlink apply autonomous orbital adjustments including station keeping and drag compensation to respond dynamically to space weather disturbances. Throughout the Gannon Storm, Starlink satellites began autonomous adjustments within hours, coordinating orbit corrections due to atmospheric drag to thousands of satellites (Foust, 2024; W. E. Parker & Linares, 2024). Sharing operational data and experiences across the satellite community further enhances collective learning and strengthens sector-wide mitigation practices (Arif, 2025).

2.5. Aviation Impact Mechanisms

Harsh radiation environments pose risks to both avionics and aviation personnel, particularly during long-duration flights and at higher altitudes where exposure to the ionosphere increases (Knipp, 2017). Ionizing radiation is an identifiable hazard in this environment and can take two forms: galactic cosmic radiation and solar energetic particle events (Phoenix et al., 2024). The NOAA SWPC conducted a recent survey underscoring the importance of better standards, procedures, and forecasting in the aviation sector (Bain et al., 2023).

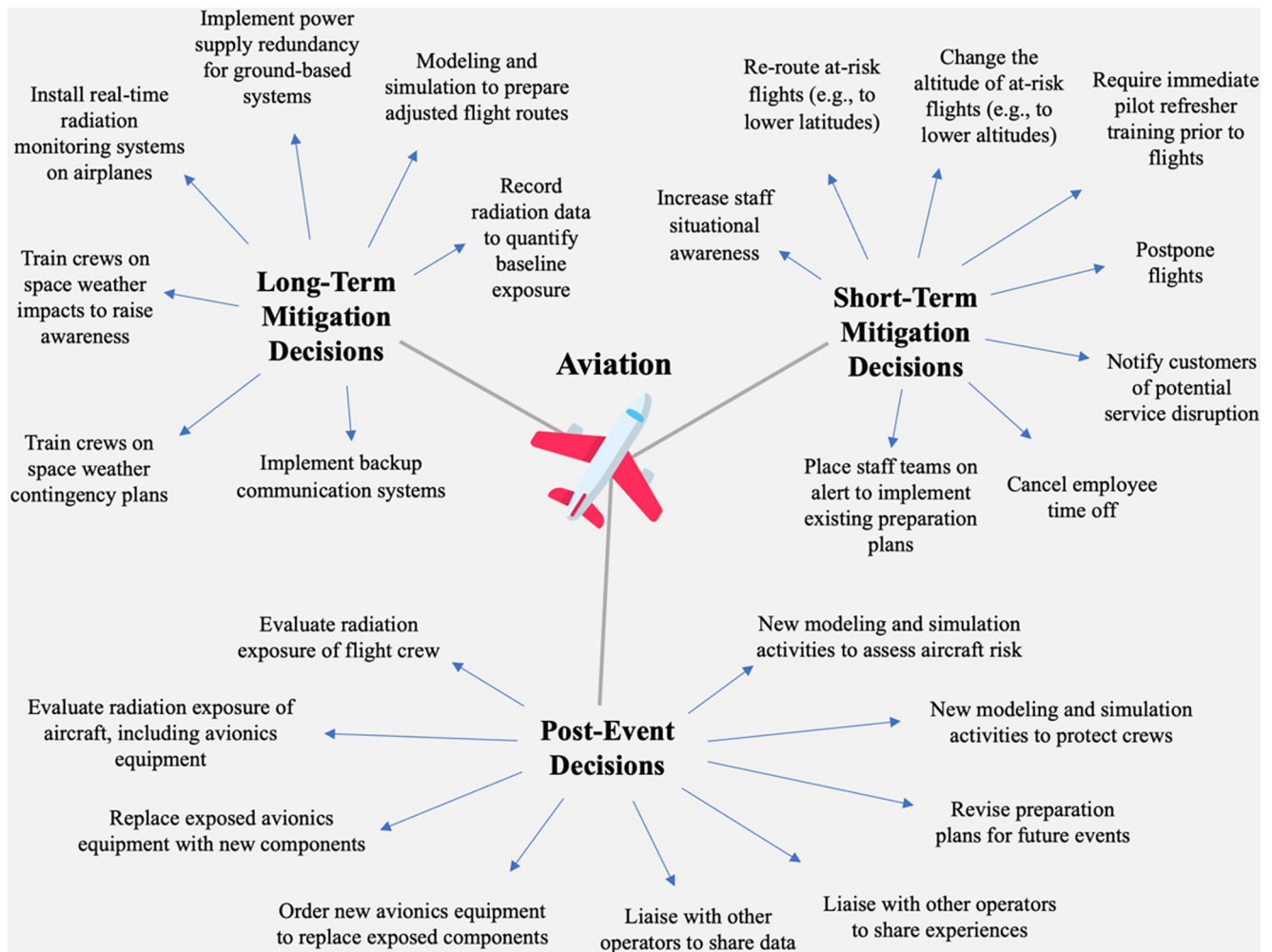


Figure 3. Strategic framework for aviation sector resilience, from workforce training and hardware redundancy (long-term) to dynamic flight path adjustments (short-term) and post-event crew and avionics exposure analysis.

There is evidence that space weather can significantly disrupt aviation operations. A study analyzing 4 million flight records found that space weather events correlated with an 81.3% increase in arrival delays and accounted for 21.5% of delays exceeding 30 min (Wang et al., 2023). Avionic electronic systems are affected by solar energetic particles that originate from space weather events (Buzulukova & Tsurutani, 2022), and because communication systems are essential to safe flight operations, disruptions to these systems may cause delays. A second study found that mean departure delay times increased by 20.7% (7.67 min) during periods affected by solar flares compared to periods without such events (Xu et al., 2023).

The Partnership for Excellence in Civil Aviation Space Weather User Services (PECASUS) authored a document summarizing space weather advisories that are given to pilots and monitors space weather conditions regarding its effect on HF Communication and satellite navigation (PECASUS, 2020). When space weather threatens these systems, the International Civil Aviation Organization (ICAO) issues advisories to flight crews (Hubert & Aubry, 2021). This PECASUS document gives further insight into current practices by which aviation professionals can be aware of possible hazards during their flight.

2.6. Aviation Mitigation Decisions

Figure 3 visualizes the range of mitigation options identified for the aviation sector. Solar radiation is an unavoidable and potent consequence of space weather, thus the ability to mitigate the radiation itself is limited (Marov, 2020). For long-term mitigation, the American National Weather Service publishes radio blackout, solar

radiation, and geomagnetic storm impact alerts that cover an 18-hr period of time (Sanders et al., 2017). Most mitigation strategies are focused on actions following a warning of a space weather event (Tezari et al., 2022). Recommended short-term measures include delaying flights when possible, to reduce time spent in irradiated atmospheric regions (Xu et al., 2023), and adjusting flight paths to lower altitudes and latitudes to minimize exposure (Schmölter & Berdermann, 2024).

The ICAO maintains a global network of space weather centers that share advisories and other information with one another, crucial in mitigation as information must be communicated to air traffic controllers worldwide to notify aircraft of new flight patterns, delays, and restrictions (Meier et al., 2020). With these forecasts and warnings, ICAO advises industry professionals to focus on areas that are known to be vulnerable to space weather events, including areas of high latitude and higher flight levels (Kauristie et al., 2021).

The Federal Aviation Administration's Risk Management Handbook (Chapter 6) outlines the hazards pilots encounter in flight and the standardized procedures for addressing them. Any type of hazard, such as degradation to communication or navigation systems, is to be dealt with through "Checklists, Standard Operating Procedures, and Best Practices," which is part of pilot initial training (FAA, 2022). The checklists and standard operating procedures are what helped create several of our mitigation strategies noted in Figure 3, as they directly correlate with mitigation procedures in-flight.

Throughout their careers, crew members are recommended to track their exposure to radiation through published dosage calculators (Phoenix et al., 2024, FAA, 2014). CARI-6, the calculator for radiation, derives the dose of galactic cosmic radiation through accounting for changes in altitude and geographic location during a flight as entered by the user. Beyond this, the FAA only mentions modeling and simulation for post-event evaluation (FAA, 2025a, 2025b). Models such as CARI-7A developed by the FAA's Civil Aerospace Medical Institute, the European Program for the Calculation of Aviation Route Doses, and the Canadian PCAIRE are all examples of modeling efforts to calculate the in-flight radiation doses for aircrews (Bottollier-Depois et al., 2009; FAA, 2025a).

3. Methodology

The review conducted identified the core mitigation techniques for reducing technological impacts during geomagnetic events from the perspective of critical infrastructure operators in the power, satellite and aviation sectors. To verify these findings and collect information on possible impacts over the solar maximum, we engaged current operators in these sectors. Gathering this data from the primary source was imperative to understand how to improve mitigation techniques for all sectors. The survey was conducted on a dedicated website (www.chrono-storm.com), which allowed operators in the power, satellite and aviation industries to anonymously complete the questions. The questions pertained to the protocols, observations, and decisions taken before, during, and after space weather events.

The survey is separated into sectors for power, satellite, and aviation operators. Once an operator selects the sector and agrees to participate, they are shown a series of questions, which include Operator Background, Long-Term Pre-Event Preparation, Short-Term Pre-Event Preparation, Operational Decisions During an Event, Post-Event Decisions, Incidents, and Interview Opportunities. The survey has multiple options for each question along with free-text boxes to record additional information. This study employed a purposive, expert-sampling approach, specifically targeting a population census of available infrastructure operators (Willie, 2024). This strategy is necessary when the "knowledgeable population" is finite and possess highly specialized technical context (Etikan et al., 2016). To maintain data integrity, representatives from 50 power utilities, 36 aviation firms, and 227 satellite organizations were invited to participate, with outreach tracked to prevent redundancy.

Participants were offered three engagement options, including (a) completion of the online survey only ($n = 22$), (b) participation in a qualitative interview only ($n = 25$), or (c) participation in both ($n = 8$). The semi-structured interviews allowed for a greater depth of contextual understanding from on-the-ground professionals, particularly regarding key events such as the May 2024 "Gannon Storm." Given the sensitivity of Critical Energy Infrastructure Information (CEII), anonymity was a primary design requirement. Following the Federal Energy Infrastructure Information (FERC) guidelines for handling CEII, no video recordings were stored, and transcripts were scrubbed of identifying technical or geographical markers to protect the security of participating networks (FERC, 2020, 2025).

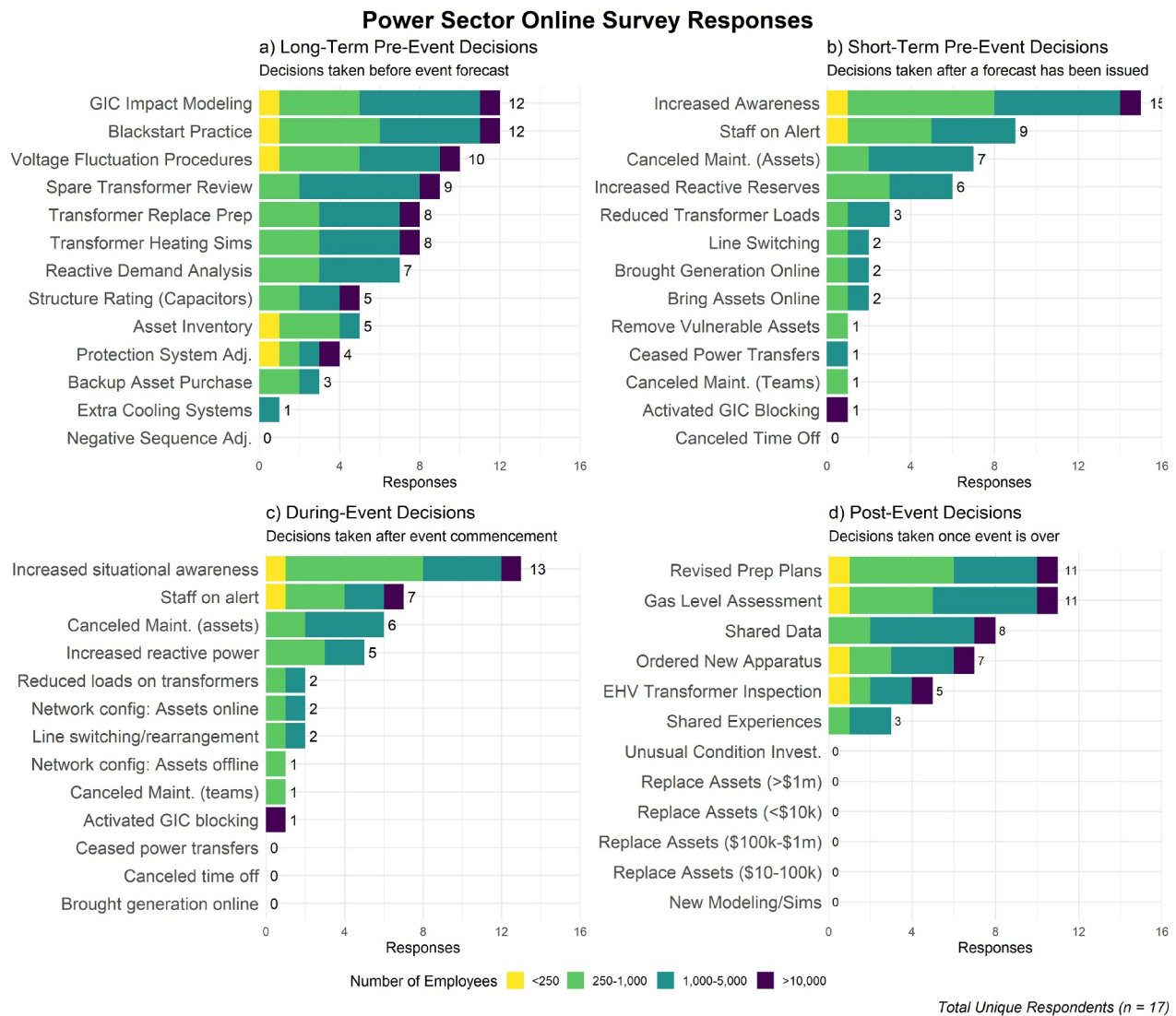


Figure 4. Mitigation decision-making in the power sector across four event phases. Data represents survey responses from 17 unique individuals, categorized by organizational size (number of employees) and the specific timing of the decision-making process relative to a space weather event.

4. Results

This section reports the results obtained from the survey and interview activities. Percentages within each subsection are based on the total respondents for that specific sector: 28 power sector respondents, 26 satellite sector respondents, and 8 aviation sector respondents. Information from each interview can be found within the supplemental information and cited with their ID number.

4.1. Power Sector

Figures 4 and 5 present the online survey and interview responses, respectively, from power grid operators regarding mitigation strategies undertaken in anticipation of potential GMDs. The horizontal bar chart categorizes responses based on the size of the organization, measured by the number of employees, to illustrate variations in preparedness strategies across different sized power utilities. The mitigation measures listed in the figure reflect a range of proactive decisions aimed at minimizing the impact of GICs on critical grid infrastructure.

Long-term mitigation strategies (Subplot A in Figures 4 and 5), primarily emphasize operational resilience through 'black-start' drills (43%) and system simulations (67%). While these activities are often space-weather

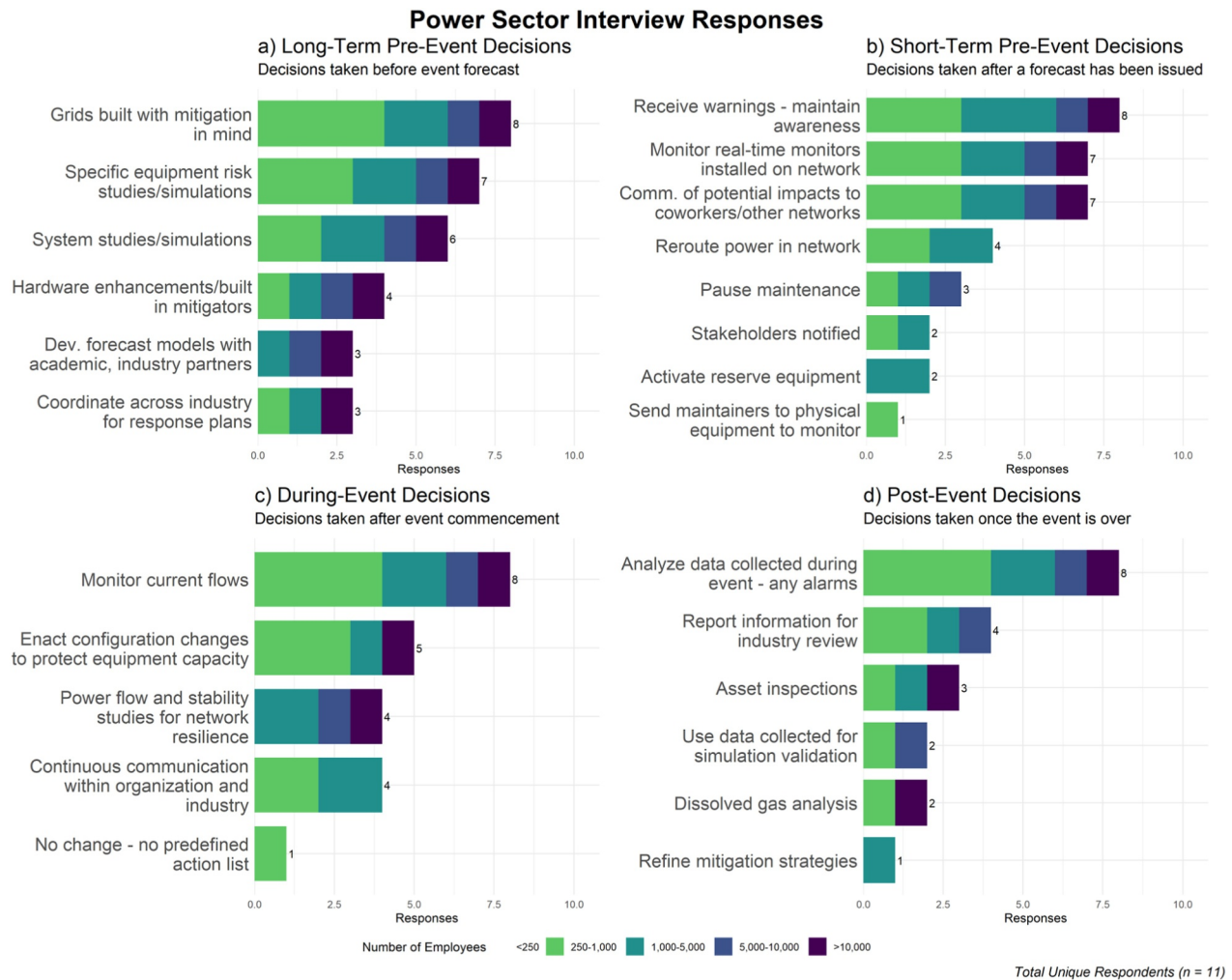


Figure 5. Mitigation decision-making in the power sector across four event phases. Data represents interview responses from 11 unique individuals, categorized by organizational size (number of employees) and the specific timing of the decision-making process relative to a space weather event.

agnostic, they reflect priority of power restoration following any system failure. Similarly, nine organizations reported having reviewed the type, condition, and location of standby spare transformers (32%), as well as seven utilizing past data to understand the reactive demand placed on transformers by GICs (25%), showing moderate engagement with asset-specific protection. Responses indicate a lower level of engagement with strategies such as purchasing backup assets, adding extra cooling systems, or re-adjusting protections, suggesting that some operators may prioritize physical asset readiness over costly physical protection system adjustments.

Notably, responses vary by organization size, with larger entities frequently engaging in modeling and simulation activities to understand GIC impacts on their networks. The correlation between organizational size and modeling activity likely reflects both resource availability and regulatory compliance. Larger entities operating high-voltage infrastructure within the United States are mandated by NERC Reliability Standard TPL-007 to conduct GMD vulnerability assessments, a requirement that inherently drives the adoption of sophisticated simulation tools (NERC, 2020). We note that smaller organizations (<250 employees) reported fewer instances of implementing extra cooling systems or reviewing manufacturer simulations for transformer heating due to GIC exposure. This may be due to potential resource constraints in undertaking such proactive measures, or they have small networks in regions of low exposure to GIC risk, or just that higher voltage assets are not in operation at smaller companies.

Next, Subplots B of both Figures 4 and 5 present survey responses from power grid operators detailing the short-term mitigation decisions, including pre-event decisions, approximately 12 hr to 4 days before a GMD occurs,

after notification from a space weather forecaster. The most reported short-term measure is “Increased staff situational awareness of the incoming space weather event” (82%), where 23 respondents highlighted the critical role of communication and awareness in preparing for potential grid disruptions. This proactive step ensures that operators are informed and prepared to execute contingency plans as needed. Nine operators also specifically indicated that utilities placed staff teams on alert to implement existing preparation plans (32%), reflecting the importance of mobilizing personnel in response to space weather forecasts. Operators commented on the existing warnings they received from NOAA ahead of solar events, one stating the “highest category of space weather warning, G5, does not provide sufficient detail about the storm severity which makes it challenging to determine necessary mitigation measures” (Id03). This sentiment was reflected within the results of the First National Survey of User Needs for Space Weather with summaries of sector-specific issues, including the needs for granularity of forecast products and identification of potential event impacts (Space Weather Advisory Group, 2024). When staff receive notification of the warnings from NOAA, there are high levels of information sharing of events, but no strict action protocols based on the warnings alone (Elvidge, 2024; Pechkis et al., 2025).

Another frequently cited action was the cancellation of routine maintenance activities (10 reports; 36%) or increasing power reserves (6 reports; 21%). These strategies suggest that operators understandably prioritize maximizing system availability and workforce readiness during space weather events. However, only one organization reported canceling employee time-off (4%), indicating that while operational readiness is emphasized, such measures may be considered unnecessary or impractical within the short notice period.

Several grid management decisions were also undertaken, including two respondents bringing additional generation capacity online (7%) and one case of ceased power transfers to other networks (4%). These actions indicate that while some utilities adjust power generation and flow, these measures are not as widely adopted as those focusing on personnel readiness and maintenance adjustments.

Network reconfiguration was employed twice to either bring more transformers, lines, and other assets online (7%) or a removal of vulnerable infrastructure from operation (4%). This suggests a strategic approach to balancing grid stability and asset protection, though the relatively low response counts imply that such interventions may be less commonly implemented across different organizations.

Additional mitigation measures include one operator's activation of GIC blocking devices for vulnerable transformers (4%), two implementing line switching and rearrangement (7%), and three reducing loads on transformers and other assets (11%). These technical interventions indicate that some utilities take direct steps to limit the potential impact of GICs on critical components, though they are not as widely applied as more general preparatory actions. The responses show variation by organization size, with larger utilities more likely to report implementing technical measures such as activating GIC blocking devices or reducing loads. During an interview, one operator stressed the importance that a switching sequence had in reducing the peak current over the May 2024 storm. It was explained that, had the correct sequence not been executed, the peak current in this US utility's network would have exceeded 200 A during the event, possibly causing damage (Id02). This sentiment suggests that larger organizations may have greater resources and flexibility to deploy protective strategies in response to space weather forecasts.

Subplots C in Figures 4 and 5 present survey responses of immediate actions utilities implement to mitigate the impact of GMD on grid stability. The responses are similar to those received for short-term pre-event decisions.

The survey findings indicate the most common reported action taken during an event was “Increased staff situational awareness of the incoming space weather event,” indicated by 13 operators in the online survey (46%), like short-term pre-event decisions. A significant number of responses, seven operators, also indicated that utilities placed staff teams on alert to implement existing preparation plans (25%), demonstrating a reliance on pre-established protocols to guide decision-making during an event. One organization reported canceling routine maintenance to free up maintenance teams (4%) while three bring more grid assets online (25%), indicating that ensuring infrastructure availability remains a priority even during an ongoing event. However, no organizations reported canceling employee time-off (0%) or ceasing power transfers to other networks (0%). Five operators increased reactive power reserves (18%) during the event, possibly as a means of stabilizing voltage fluctuations induced by GMDs. Additionally, two respondents engaged in network reconfiguration to bring more transformers, lines, and other assets online (7%) or take vulnerable infrastructure offline (4%), with five interviewees citing enacting configuration changes to protect equipment capacity (18%). These results indicate that while grid

adjustments are considered, they may be less frequently executed in real time due to operational constraints, with most decisions potentially being taken in advance.

As with short-term pre-event decisions, most grid operators primarily focus on maintaining situational awareness, mobilizing staff, and making selective adjustments to network maintenance and power reserves. The relatively low frequency of grid reconfiguration or protective device activation may reflect the preference for pre-event planning and proactive mitigation rather than reactive adjustments.

Finally, for the power sector both Figures 4 and 5 Subplots D presents survey responses from power grid operators regarding the actions taken after a space weather event has concluded. These post-event decisions reflect the steps utilities undertake to assess the impact of GMDs, share insights, and improve preparedness for future occurrences. A predominant response among operators was to share information, with three opting to share experiences (11%), and 11 operators sharing data to industry (39%). Data-sharing has played a key role in evaluating the effects of the May and October 2024 events, particularly to inform future risk management strategies.

Fifteen operators refined their understanding of space weather impacts on grid infrastructure by stating that they had analyzed data recorded during the events, to include additions to simulation and modeling (54%). Most operators that were interviewed agree with the survey results, stating that modeling and verification of models against recorded data helped prepare their systems for future events. Fourteen operators conducted inspections of their equipment and networks following space weather events (39%), and 12 revised their preparation plans, reflecting a proactive approach to identifying any unforeseen impacts.

Interestingly, no respondents indicated that they “Physically replaced exposed assets with new components” across any value category (>\$1M, \$100K-\$1M, \$10K-\$100K, <\$10K) (0%). This suggests that immediate asset replacement after a GMD was unnecessary so far over the 2024 solar maximum.

Overall, the results emphasize that post-event actions for recent events focused on power-sector collaboration through data sharing and analysis, with the aim of refining future preparedness plans through shared understanding. The emphasis on modeling, experience-sharing, and asset inspections suggests that utilities have been focusing on learning as much as possible from these recent space weather events, with the aim of enhancing power grid resilience against future GMDs.

4.2. Satellite Sector

Figure 6 shows the long-term decisions made proactively, prior to the issuance of specific space weather forecasts, to enhance the resilience of satellite systems in both Low and Geostationary Earth Orbits.

Among the most frequently adopted strategies, 20 respondents reported the evaluation of adequate satellite shielding thickness and ongoing quality levels (77%). Moreover, 19 anonymous operators explained that many of their satellites are over-designed for solar radiation and that the radiation engineers personalize the protection to the mission and orbit, not just space weather events (73%). Furthermore, specifically within the survey responses, modeling and simulation approaches were widely employed, 10 instances of quantifying expected radiation exposure in satellite system design (77%) and nine for simulating orbital trajectories to assess cumulative radiation exposure (69%). One interviewee stated that “it is all about design” when asked about their operators’ long-term preparation for space weather (Id18). Other interviewees explained that not much can be done after launch, which requires the design process to have five to 10 years as the “long-term” mitigation period (Id13, Id17).

Redundancy in satellite system design also emerged as a key mitigation strategy. Sixteen respondents indicated that they had developed redundancy design for onboard systems to safeguard against radiation exposure (62%), ensuring that critical components maintain operational functionality in adverse space weather conditions. Another strategy involved monitoring satellite asset orbital trajectories to predict which assets would experience the largest cumulative radiation exposure (8 respondents, 31%), enabling operators to implement informed risk mitigation measures.

Ground-based infrastructure was specifically considered by seven operators within the long-term mitigation framework (27%). Power supply redundancy for Earth stations was implemented ensuring continued operability of control systems in the event of space weather-induced disruptions. Similarly, optical fiber link redundancy was established by five operators to enhance communication network resilience (19%), mitigating the risks of ionospheric disturbances and signal degradation.

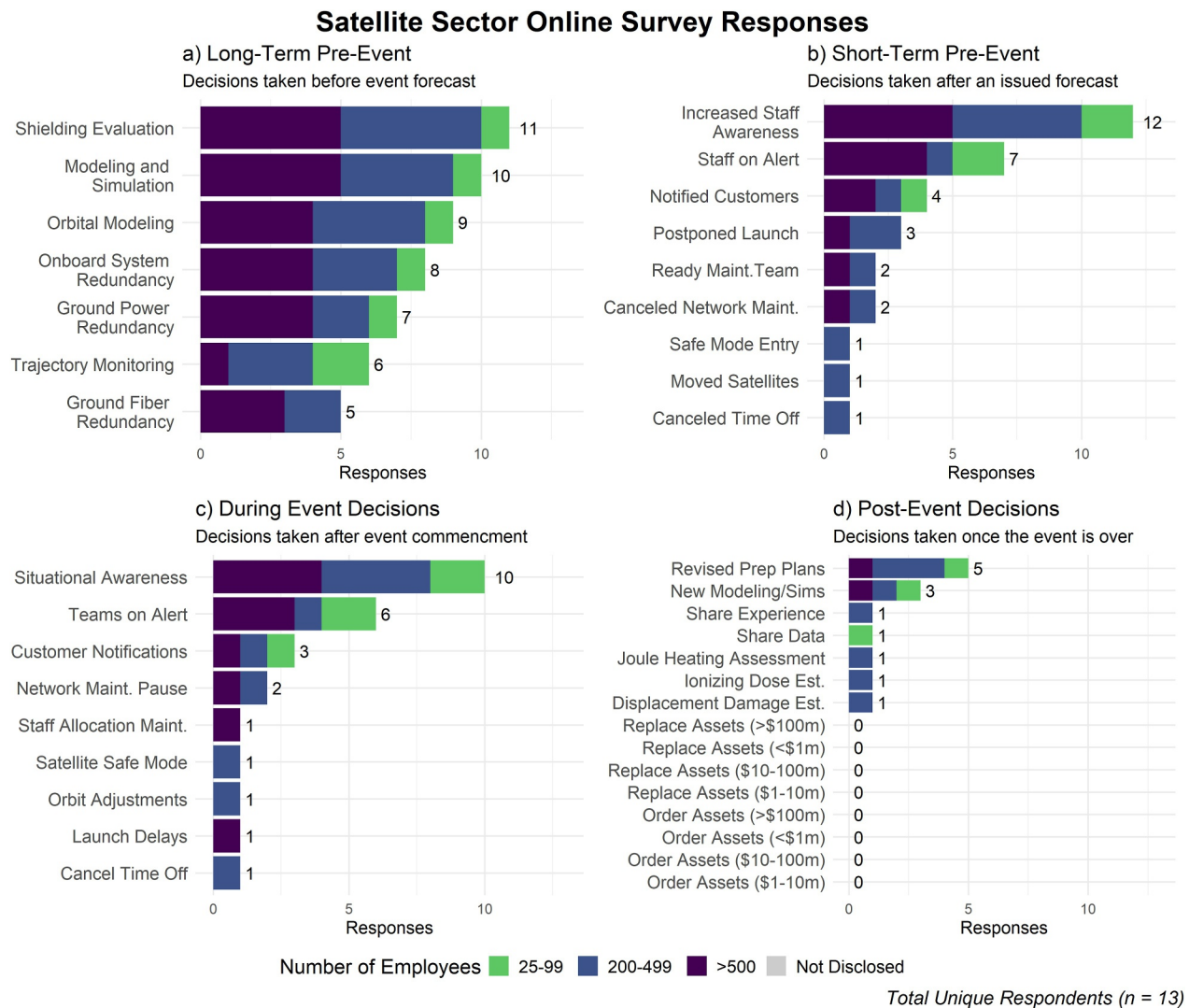


Figure 6. Mitigation decision-making in the satellite sector across four event phases. Data represents survey responses from 13 unique individuals, categorized by organizational size (number of employees) and the specific timing of the decision-making process relative to a space weather event.

Short-term operational decisions undertaken by satellite operators are presented in Subplot B of Figures 6 and 7, in response to space weather forecasts indicating an imminent GMD. Among the most frequently reported responses, increasing staff situational awareness of the impending space weather event was reported by 18 operators (69%). This highlights the emphasis placed on ensuring personnel are prepared to monitor system performance and execute contingency plans as conditions evolve. Additionally, seven of the survey respondents indicated they placed staff teams on alert to implement established preparation protocols was another widely adopted strategy (27%). During an interview, one operator explained the importance of shared understanding during events, stating that out of the history of operations during storms, one kinetic response occurred: moving a satellite to another orbit (Id15).

Communication strategies were also employed as part of the short-term response framework. Eight operators notified customers and stakeholders about the upcoming event and the potential for service disruptions (31%). This measure manages expectations and ensures that end-users and mission stakeholders are aware of possible performance impacts.

Several operators adjusted their operational schedules in anticipation of heightened space weather activity. One respondent postponed launch activity for new satellites (4%), minimizing the risk of exposing newly deployed assets to elevated radiation levels or increased atmospheric drag. Additionally, one operator canceled routine

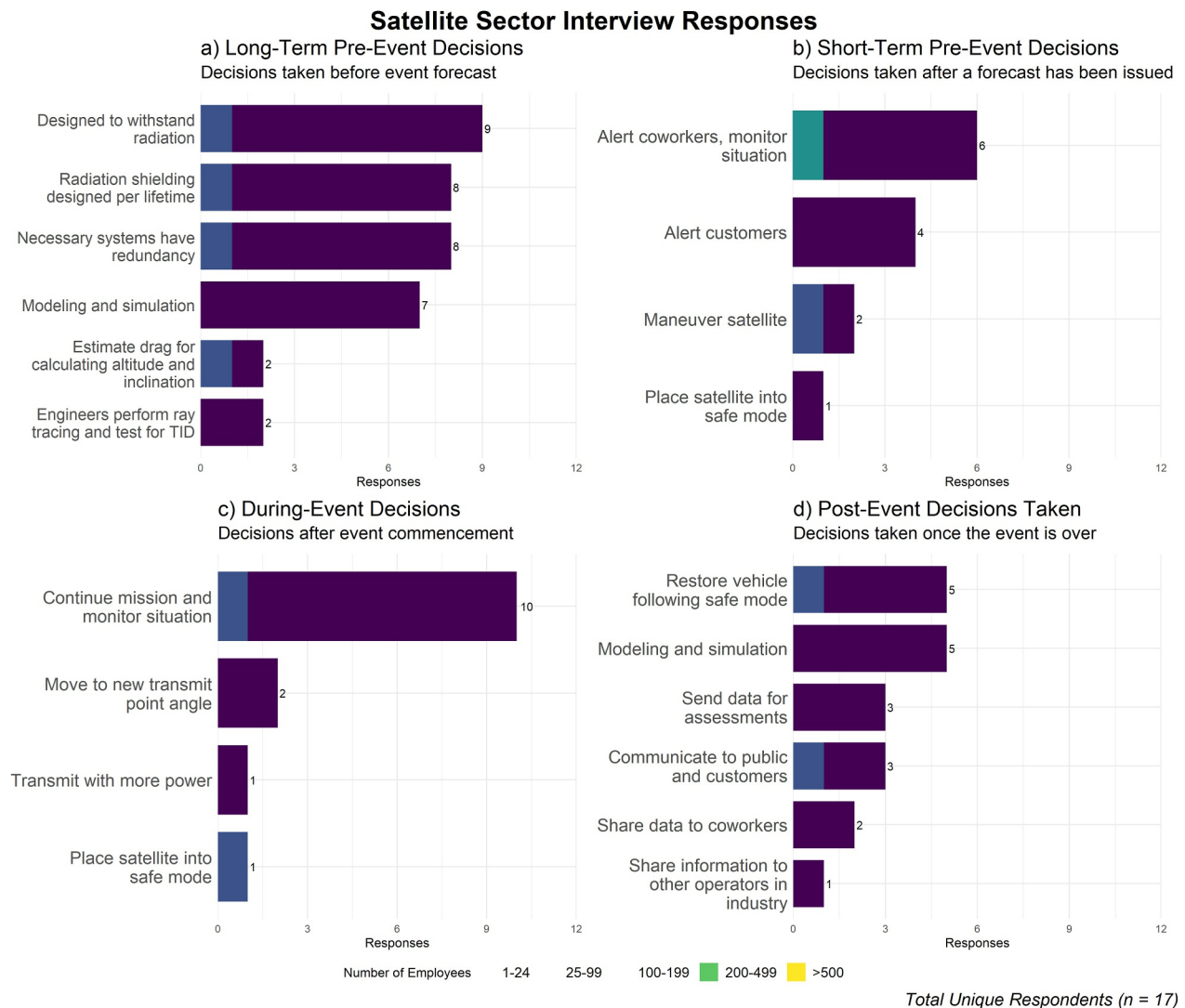


Figure 7. Mitigation decision-making in the satellite sector across four event phases. Data represents interview responses from 13 unique individuals, categorized by organizational size (number of employees) and the specific timing of the decision-making process relative to a space weather event.

network maintenance to free up teams for more immediate operational tasks (4%) and two suspended routine maintenance of terrestrial assets (8%), likely to reduce exposure to space weather-induced power grid fluctuations or ionospheric disturbances affecting ground communications.

Protective actions for satellite assets were less commonly reported but still noteworthy. One operator moved at-risk satellites out of expected orbit to mitigate exposure to hazardous conditions (4%), while two placed satellites into safe mode (8%), a precautionary step to protect onboard electronics from space weather induced anomalies. One anonymous operator explained why they avoided placing the satellites into safe mode, citing that the loss of services would outweigh the risk of damage, and that the customers almost always opted for continued service (Id19).

Subplot C in Figures 6 and 7 presents the immediate operational responses undertaken by satellite operators once a space weather event actively impacts Earth's near-space environment. These actions reflect adaptations to mitigate disruptions and protect critical satellite infrastructure during geomagnetic storms.

The most widely reported measure during this phase with 10 reports involved enhancing staff situational awareness of the unfolding space weather event (38%). Six operators placed their teams on heightened alert to execute pre-existing contingency protocols (23%), ensuring that personnel were prepared to intervene as

necessary. Communication with external stakeholders remained an important aspect of operational decision-making, where three operators notified customers and stakeholders of the ongoing event and potential disruptions to services (12%).

Subplots D of Figures 6 and 7 illustrate the post-event decisions taken by satellite operators after a space weather event had concluded. These actions reflect efforts to assess the impacts of the disturbance, refine mitigation strategies, and enhance resilience for future solar activity. Unlike the immediate responses observed during the event, post-event decision-making is characterized by retrospective analysis, collaborative engagement, and strategic planning.

The primary focus of post-event activities was the revision of preparation plans, with five of the survey operators reporting updates to their mitigation strategies based on observed system performance and response effectiveness (19%). This underscores the iterative nature of space weather resilience, where each event provides an opportunity to refine protocols and strengthen operational preparedness. Additionally, eight operators undertook new modeling and simulation activities, likely aimed at improving predictive capabilities and refining spacecraft vulnerability assessments (31%). In comparison, one satellite operator explained in an interview that the specifications used were over a decade old, losing relevancy and operator trust for accuracy (Id16).

Post-event technical assessments were also conducted to evaluate the impact of the space weather disturbance on satellite assets and operational infrastructure. One operator assessed Joule heating effects, a critical factor influencing atmospheric drag and satellite orbital decay (4%), while another estimated the TID experienced by spacecraft components, which can degrade electronic performance over time (4%). A further respondent evaluated DDD, an important metric for understanding long-term material degradation due to charged particle interactions (4%).

Collaborative efforts among operators were evident, albeit limited in scale. Three respondents liaised with other operators to share impact data (12%), while two others engaged in discussions to exchange operational experiences (8%). Such collaboration plays a vital role in advancing industry-wide best practices for space weather mitigation, particularly given the interconnected nature of satellite-based services.

Notably, no operators reported taking direct asset replacement actions in response to the event. No respondents indicated having physically replaced satellites or placed orders for new spacecraft to compensate for exposure-related degradation. Four operators did indicate that they did restore the satellites in safe mode (19%), suggesting that, at least for this solar maximum, operators did not experience immediate catastrophic failures necessitating asset replacement.

4.3. Aviation Sector

Figure 8 shows all mitigation decisions made across all stages of an event for our aviation respondents. Long-term pre-event decisions, shown in Subplot A of Figure 8, were mostly focused on training, with five annual (63%) and one initial (13%) training responses, reporting both having portions focused on general weather with a smaller portion on space weather impacts. One interviewee specifically mentioned that dispatch also has an annual training that includes space weather (13%). Two operators also reported forming space weather teams (25%). Responses indicate an emphasis on training ahead of any event, similar to both the power and satellite sector.

Short-term pre-event decisions, shown in Subplot B of Figure 8, had consistent results pointing toward shared understanding of the alerts published. Five pilots reported that they would receive alerts within their flight packets (63%), three alerted by dispatch during pre-flight briefings (38%), and four responded they would continue to have access to the information on their flight iPads (50%). One respondent reported that they would run crew rehearsals/drills on notification of an alert (13%).

Interviewees were focused primarily on the continuation of mission during the event phase of a storm (Subplot C of Figure 8), with no report of delays as recommended by literature. In fact, four responded to continuing mission with no change (50%), six reported that HF communications were impacted, then switched to satellite radios to contact dispatch (75%), and two collected data during the event (25%). Interestingly, three pilots reported changes in altitude due to an enlarged ozone layer (38%), aligning with literature recommendations. Two respondents talked about carrying more fuel in response to the alerts and events (25%). This underscores the importance of retaining flexibility in actions to hazards and open communication lines to complete flights as planned.

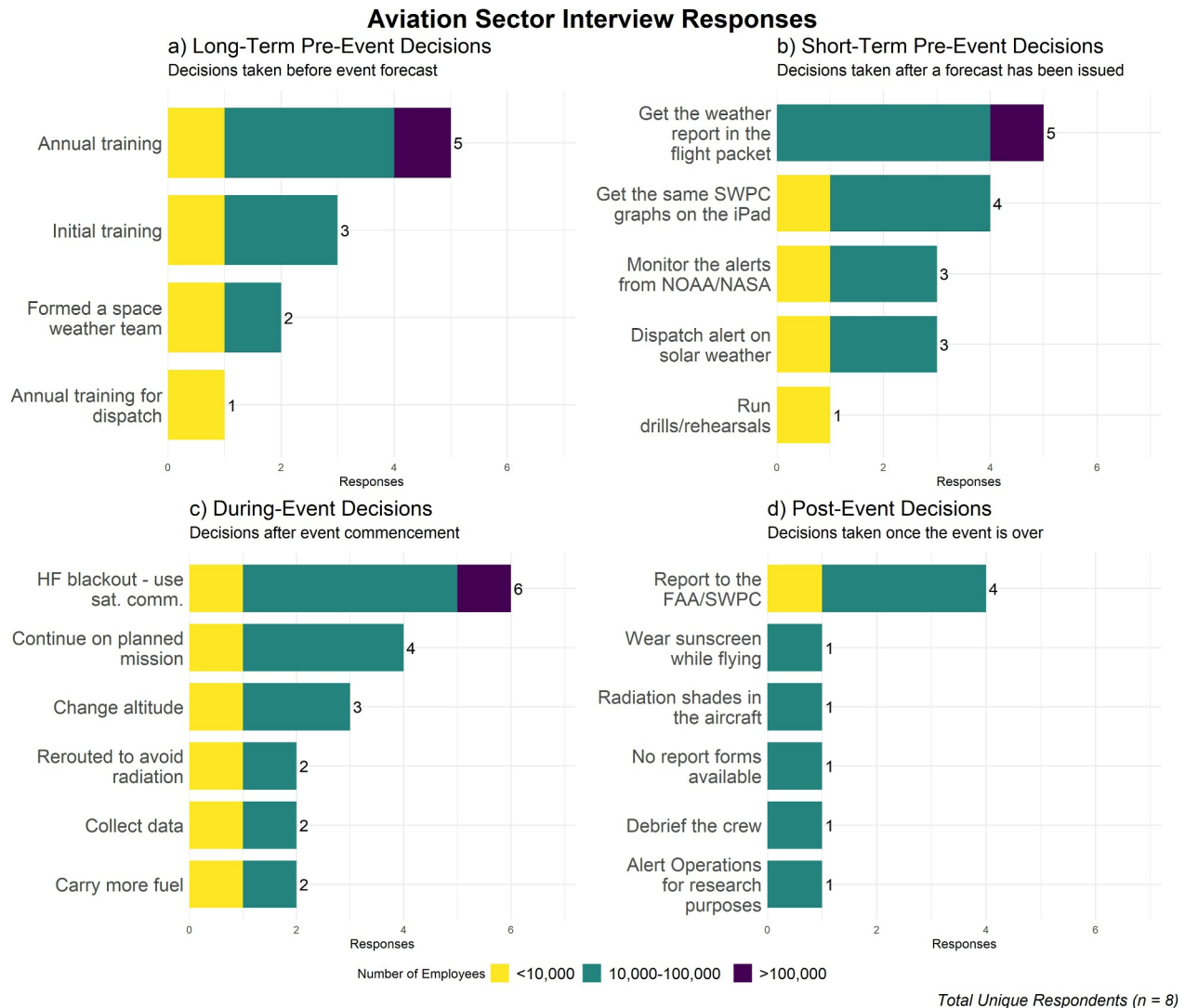


Figure 8. Mitigation decision-making in the aviation sector across four event phases. Data represents interview responses from 8 unique individuals, categorized by organizational size (number of employees) and the specific timing of the decision-making process relative to a space weather event.

Post-event decisions (Subplot D of Figure 8) contained actions centered mostly on alerting others to what happened and debriefing crews. The only statement to have consensus was four reports of impacts back to the FAA and SWPC (50%), with one mention of not having the ability to report but the want to do so (13%). Two pilots reported changes in preparation and aircraft, with inclusions of radiation shielding into aircraft (13%) and of increased use of sunscreen (13%).

Overall, six interviewees reported instances of HF communication interruptions (75%), three reported GPS impacts (38%), and two changes to routing of flights (25%), suggesting that space weather has a play into mission planning and execution. Education is key within the aviation sector, dominating long-term pre-event and short-term pre-event phases with training and shared information. For example, one respondent shared that they would notify any pregnant crew members or pilots of the risks of flying during an event due to increased levels of radiation.

5. Discussion

Across all sectors, operators implemented 67 out of 91 identified mitigation options (74%) during Solar Cycle 25. This discussion interprets these patterns relative to the three research questions, comparing literature-recommended practices with actual behaviors during the May and October 2024 storms.

The first research question was to identify the main mitigation decisions used in the three critical infrastructure sectors of interest, across four different stages of a space weather event (long term pre-event, short term pre-event, during event, post-event).

While some literature emphasizes capital intensive physical hardening, such as EHV transformer redundancies or GIC blocking devices, operational reality is driven by cost-efficiency. Given only 4% of respondents (1 of 28) reported activating GIC blocking devices during an event suggest there are few installations because of their high cost and the careful planning needed to ensure blocking in one part of a network does not increase the exposure risk in another part. No respondents reported changes to the protection systems. Instead, the power sector relies on preparatory “soft” mitigation: 43% (12 of 28) practiced black-start procedures and 64% (18 of 28) conducted simulations, consistent with preparation for all system events. This suggests that rather than preventing GIC entry through expensive hardware, operators prioritize swift restoration over absolute prevention. The mission continuity prioritized by operators is a calculated financial decision; immediate revenue loss of network reconfiguration outweighs the possible high cost of EHV equipment replacement.

The satellite industry faces the physical inaccessibility of orbital assets that shifts the mitigation burden almost entirely to the pre-launch phase. Our data shows that 77% (20 of 26) of operators focus on shielding thickness and 62% (16 of 26) on system redundancy. This aligns with the long-term focus identified in literature, but our results show a difference during the event phase. While literature strongly recommends placing the assets into safe mode, or orbital maneuvering to mitigate radiation, only 8% (2 of 26) of respondents implemented these measures. This gap is critical: operators reported that the loss of services often outweighs the perceived risk of hardware damage (Id19). As autonomous orbit maneuvers are adopted by more operators, the ability to mitigate risk during and following events will increase. Furthermore, the post-event phase in the satellite sector suffers from a unique data perishability. Unlike terrestrial sectors, failed assets cannot be physically recovered for forensic analysis once they are decommissioned or re-entered into the atmosphere, limiting the industry's ability to refine models based on actual hardware degradation. Aviation mitigation is characterized by tactical flexibility rather than strategic avoidance. Despite literature focusing on flight path diversions decided after a forecast advisory, our interviews revealed that 50% (4 of 8) of pilots continued missions with no change, or continued on missions diverted by dispatch before the event start (Horne et al., 2026). However, resilience was demonstrated through system-switching: 75% (6 of 8) reported HF communication impacts but successfully transitioned to satellite-based communication. This suggests that for aviation, mitigation is not the avoidance of hazards, but the existence of redundant communication layers that allow the mission to proceed. Research question two evaluated which mitigation decisions were actively implemented during the peak of Solar Cycle 25, specifically during the powerful storms of May and October 2024. While awareness was high across all sectors, our results suggest a significant resource-driven disparity in preparedness. In the power sector, larger organizations (>250 employees) were more likely to engage in modeling and simulation (64% vs. 18% in smaller firms). This suggests that while space weather is a global phenomenon, the capacity to respond is stratified. Larger utilities, often subject to NERC TPL-007 mandates, have integrated space weather into their system-wide analysis, whereas smaller organizations reported fewer instances of proactive measures like implementing extra cooling systems or reviewing manufacturer simulations.

A finding of this study is that, despite the severity of the May 2024 Gannon Storm, no respondents reported immediate large-scale purchases, repairs, or physical changes to their networks. This lack of “hardware trauma” suggests that current design margins were sufficient for these events, or that impacts are cumulative rather than catastrophic. This is supported by post-event actions, where the focus shifted from physical repair to data analysis; 54% (15 of 28) of power operators and 31% (8 of 26) of satellite operators conducted new modeling and simulation activities following the events to refine their vulnerability assessments.

In the satellite sector, the emphasis remained almost entirely on preparatory design, reflecting the reality that operators cannot easily modify hardware once in orbit. For the aviation sector, the primary during-event decision was information sharing; contrasted with only 38% (3 of 8) pilots reporting change of altitude in route, the success

of these missions relied on updated training and notification systems. This underscores that for all sizes of organizations, success during a solar maximum is less about immediate hardware changes and more about the robustness of the information-sharing ecosystem before and after the event.

Regarding research question three, a difference was observed between the prescriptive measures found in academic literature and the operational practices of critical infrastructure providers. While literature advocates physical interventions such as GIC blocking devices for power grids or orbital maneuvering for satellites, industry adoption of these measures remains low, suggesting the need for more evidence of damage before investing time and capital into physical mitigation options. Our data shows that operators implemented 74% of identified strategies, but these were the “lower-cost” options such as increased situational awareness (82% or 23 of 28 in power, 69% or 18 of 26 in satellite) and modeling.

A core reason for this difference is the difficulty in attribution. Though not directly addressed through a survey question, respondents revealed during the qualitative interviews that operators often cannot definitively isolate GIC-induced damage from routine environmental stress. As noted by respondents, GMD may not be the sole cause of failure but rather the final push for an asset already nearing a hazard state. This creates a benefit-uncertainty barrier; without clear proof that expensive hardware replacements (such as EHV transformers) are solely necessitated by space weather, firms are hesitant to commit capital. This sector-specific hesitation is also tied to forecast granularity. One operator noted that even a G5 warning “does not provide sufficient detail about the storm severity,” making technical mitigation challenging (Id03). This aligns with the 2024 National Survey of User Needs (Space Weather Advisory Group, 2024) highlighting the need for impact-based forecasting. Consequently, the power sector relies on “soft” measures; increased situational awareness was the most popular strategy, though industries did align with literature on actions such as increasing reactive reserves (18% or 5 of 28 survey respondents) and decreasing loading on susceptible equipment (11%, 2 of 28).

In the satellite industry, both literature and practices aligned on long-term mitigation, with a heavy focus on shielding and redundancy (62%, 16 of 26). The literature recommends risk-adverse approaches, such as transitioning assets into low-power safe modes (Arif, 2025; Qiao et al., 2024; Trimoreau, 2022) or executing orbital maneuvers based on exposure predictions (Miteva et al., 2023; Spence et al., 2022), but most operators in this study avoided these measures. Respondents indicated that they cannot justify the resulting loss of service and revenue, suggesting that economic viability often supersedes the theoretical safety protocols.

Finally, while the aviation sector results are limited in sample size, it demonstrated the strongest alignment with literature regarding human-centric mitigation. Decisions remain pilot-centric, with training and awareness dominating the strategy. The report of pregnant crew members avoiding flights (Id28) underscores that industry is successfully translating literature regarding radiation risks into behavioral changes, even if technical flight diversions remain less frequent.

One benefit of collecting and recording perishable data immediately following the 2024 events is the empirical insight gained into how critical infrastructure is protected in practice. Currently, a mismatch exists between the policy-level perception and operational reality. While the National Space Weather Strategy and Action Plan, directed by the SWORM Subcommittee (SWORM, 2023), seeks to ensure national security through infrastructure resilience, our findings suggest that industry's path to resilience is primarily through data-driven agility rather than static hardware upgrades.

Governing bodies have a vested interest in updating models and facilitating knowledge sharing. However, while literature focuses on physical system changes, industry participants expressed a critical need for precise, impact-based modeling and more granular weather warnings. For many operators, a change in physical equipment is not economically practical; conversely, targeted dynamic predictions from NOAA would provide the high-impact, low-cost mitigation they require to make informed operational calls during G5-class events.

While this study provides a unique window into the 2024 solar maximum, several limitations remain. Despite the difficulty collecting sensitive operational data from organizations, this study had a response rate of 18% with a sample of 55 unique respondents, while the sample size was sufficient for addressing our research questions, the response rate reflects the challenges of gathering data on sensitive operational actions. The limited participation from certain regions and government assets is likely due to the unclassified nature of the study and corporate concerns regarding the sensitivity of infrastructure vulnerabilities. Future research should expand this perishable

data set by incorporating other GNSS users (e.g., precision agriculture and maritime), who represent a growing but understudied segment of space weather stakeholders.

6. Conclusion

This study captures a critical inflection point in terrestrial and orbital resilience during the peak of Solar Cycle 25. By contacting 313 operators and analyzing 55 unique responses (18% response rate), we found that while industry enacted 74% of 91 identified mitigation options, these actions skewed heavily toward low-cost measures, such as situational awareness and modeling rather than capital-intensive hardware interventions. Operators across all three sectors revealed that they are willing to accept the risk of potential space weather impacts to maintain service continuity and therefore revenue. The findings underscore a fundamental mismatch between idealized recommendations and what takes place operationally by private critical infrastructure operators with constrained resources and profit maximizing objectives. The most consistent cross-sector finding is that operators need precise, impact-based, localized event forecasts to make informed mitigation decisions. Future research should quantify the cost-benefit ratios for the range of mitigation options identified here. Ultimately, this study provides new empirical information to help us understand how actual critical infrastructure operators are addressing space weather risks in Solar Cycle 25, which shows a substantial difference from what we might have expected based on academic sources alone.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The text data gathered from the interviews in the study are available at Zenodo via <https://doi.org/10.5281/zenodo.19099439> with Creative Commons Attribution 4.0 International license (LaNeve et al., 2026). The R scripts and online survey data csv files used to execute/report on the analyses in the paper can be found at <https://github.com/CaitlinLaNeve/SpaceWeatherMitigation> (CRAN, 2026) and is preserved at <https://doi.org/10.5281/zenodo.19098930> (LaNeve & Oughton, 2026).

Acknowledgments

We gratefully acknowledge support from the US National Science Foundation grant entitled RAPID: Collecting Perishable Critical Infrastructure Operational Data for May 2024 Space Weather Events (2434136). This grant is co-funded by the GEO/AGS Space Weather Research program and the ENG/CMMI Humans, Disasters, and the Built Environment (HDBE) program.

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